CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2015 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Define Database Management System (DBMS).

Definition: A DBMS is software designed to manage, store, query, and maintain structured data securely and reliably, guaranteeing data independence and concurrent multi-user access.

Q2 (3M) Differentiate between Schema and Instance.

Comparison: A Schema is the overall structural blueprint designed once, while an Instance is the actual operational collection of data tuples present at any given moment.

Q3 (3M) What is Data Independence?

Definition: The capacity to change the data schema at one architectural level without altering schemas at higher levels.

Q4 (3M) What do you understand by Hashing? Explain with example.

Definition: Direct access technique mapping search keys to physical bucket addresses via hash function h(k) = k \pmod m for O(1) lookups.

Q5 (3M) What are Nested SQL Queries? Explain with example.

Definition: A query enclosed inside another SQL query, passed via `WHERE` or `HAVING` clauses using operators like `IN`, `EXISTS`, `ALL`, `ANY`.

```
SELECT EmpName FROM Employee
WHERE Salary > (SELECT AVG(Salary) FROM Employee);
```

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain the Three-Level Architecture of DBMS and its primary objectives.

Architecture: View Level → Conceptual Level → Internal Storage Level. Primary objective is achieving program-data independence and providing customized views to multiple users.

Q7 (7.5M) Explain BCNF (Boyce-Codd Normal Form) and compare with 3NF.

Explanation: BCNF requires every determinant in a non-trivial functional dependency to be a super key. Stricter than 3NF, eliminating all redundancy based on functional dependencies.

Q8 (7.5M) Explain Concurrency Control and ACID properties.

Explanation: Mechanisms ensuring Atomicity, Consistency, Isolation, and Durability when multiple transactions access the database simultaneously.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain ER Modeling: Entities, Attributes, Relationships, and Cardinality Constraints. Draw complete ER diagram for College Administration.**

Full Analysis: Complete conceptual ER model with Student, Faculty, Department, and Course entity sets, relationships, and cardinalities.

Q10 (15M) **Define Normalization. Why do we need to normalize? Explain 1NF, 2NF, and 3NF with examples.**

Full Normalization Process: Comprehensive breakdown of insertion, deletion, and update anomalies, followed by step-by-step normalization through 1NF, 2NF, and 3NF.

Q11 (15M) **Explain Relational Algebra Operations with expressions and SQL equivalents.**

Operations: Selection (\sigma), Projection (\pi), Cartesian Product (\times), Union (\cup), Set Difference (-), and Natural Join (\bowtie).

Q12 (15M) **Explain Indexing and Hashing: Primary vs Secondary Index, Dense vs Sparse Index.**

Indexing Schemes: Clustered/Primary indexing on ordered files vs Non-clustered/Secondary indexing on unordered candidate keys, dense vs sparse index trade-offs.

Q13 (15M) **Explain Log-Based Recovery and Checkpoints.**

Recovery: Detailed explanation of Undo/Redo protocols, Write-Ahead Logging (WAL), system checkpoints, and crash restart recovery algorithms.
